

BASH SCRIPTING – 01

1) Launched an EC2 Instance and connected it to Linux machine using ssh command.

```
salm@SALMANBABSAIL MINGW64 ~/OneDrive/Desktop
$ ssh -i Linux_test1.pem ec2-user@54.175.188.142
The authenticity of host '54.175.188.142 (54.175.188.142)' can't be established.
ED25519 key fingerprint is SHA256:DFFmdIs6/u2V4x40qxBVig6imgXJictNgb8y1wHFoog.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.175.188.142' (ED25519) to the list of known hosts
```



```
'_#_
~\####_      Amazon Linux 2023
~~\#####\
~~\###|
~~\#/       https://aws.amazon.com/linux/amazon-linux-2023
~~V~'_->
~~~~
~~~.~.~
~/m/'_>
```

```
[ec2-user@ip-172-31-33-47 ~]$ |
```

2)Created a test bash script named Hello World

```
salm@SALMANABSAIL MINGW64 ~/OneDrive/Desktop
$ ssh -i Linux_test1.pem ec2-user@54.175.188.142
The authenticity of host '54.175.188.142 (54.175.188.142)' can't be established.
ED25519 key fingerprint is SHA256:DFFmdIs6/u2V4x40qxBVig6imgXJictNgb8y1wHFoog.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.175.188.142' (ED25519) to the list of known hosts
```

```
      #_
     / \   ###_       Amazon Linux 2023
    ~~~~ \#####\
         \####|
         \#/
        V~'_-'>      https://aws.amazon.com/linux/amazon-linux-2023
           ^
          / \
         /   \
        /     \
       /       \
      /         \
     /           \
    /             \
   /               \
  /                 \
 /                   \
/_/m/'_-'>
```

```
[ec2-user@ip-172-31-33-47 ~]$ |
```

```
root@ip-172-31-33-47:/salman/scripts
[root@ip-172-31-33-47 scripts]# vi Test.bash
[root@ip-172-31-33-47 scripts]# chmod 755 Test.bash
[root@ip-172-31-33-47 scripts]# cat Test.bash
#!/bin/bash

a='Hello world'

echo
echo $a
echo
[root@ip-172-31-33-47 scripts]# ./Test.bash

Hello world

[root@ip-172-31-33-47 scripts]# |
```

3) Ran a script with process commands.

```
root@ip-172-31-33-47:/salman/scripts
[root@ip-172-31-33-47 scripts]# vi commands.bash
[root@ip-172-31-33-47 scripts]# chmod 755 commands.bash
[root@ip-172-31-33-47 scripts]# cat commands.bash
#!/bin/bash

free -m
pwd
ps -ef

[root@ip-172-31-33-47 scripts]# ./commands.bash
              total        used        free      shared  buff/cache   available
Mem:           912          161          515           0         235         619
Swap:           0           0           0
/salman/scripts
  UID          PID    PPID    C  STIME TTY          TIME CMD
root           1        0    0  09:52 ?        00:00:01 /usr/lib/systemd/systemd --switched-root
root           2        0    0  09:52 ?        00:00:00 [kthreadd]
root           3        2    0  09:52 ?        00:00:00 [pool_workqueue_release]

root      2099      2098    0  09:55 pts/1        00:00:00 su
root      2100      2099    0  09:55 pts/1        00:00:00 bash
root      2339         2    0 10:03 ?          00:00:00 [kworker/1:0-events]
root      2394         2    0 10:05 ?          00:00:00 [kworker/0:1-xfs-reclaim/nvme0n1p1]
root      2560         2    0 10:10 ?          00:00:00 [kworker/1:1-cgroup_free]
root      2563         2    0 10:10 ?          00:00:00 [kworker/0:0-cgroup_free]
root      2675      2007    0 10:14 ?          00:00:00 systemd-userwork: waiting...
root      2676      2007    0 10:14 ?          00:00:00 systemd-userwork: waiting...
root      2677      2007    0 10:14 ?          00:00:00 systemd-userwork: waiting...
root      2780         2    0 10:17 ?          00:00:00 [kworker/0:2]
root      2783      2100    0 10:18 pts/1        00:00:00 /bin/bash ./commands.bash
root      2785      2783    0 10:18 pts/1        00:00:00 ps -ef
[root@ip-172-31-33-47 scripts]#
```

4) Ran a bash script with more commands and created new files in the same script.

```
root@ip-172-31-33-47:/salman/scripts
[root@ip-172-31-33-47 scripts]# vi Test2.bash
[root@ip-172-31-33-47 scripts]# chmod Test2.bash
chmod: missing operand after 'Test2.bash'
Try 'chmod --help' for more information.
[root@ip-172-31-33-47 scripts]# chmod 755 Test2.bash
[root@ip-172-31-33-47 scripts]# cat Test2.bash
#!/bin/bash

date
cal
whoami
touch a b c d

[root@ip-172-31-33-47 scripts]# ./Test2.bash
Fri Jul 3 10:23:12 UTC 2026
      July 2026
Su Mo Tu We Th Fr Sa
      1  2  3  4
  5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31

root
[root@ip-172-31-33-47 scripts]#
```

```

root@ip-172-31-33-47:/salman/scripts
[root@ip-172-31-33-47 scripts]# vi admin.bash
[root@ip-172-31-33-47 scripts]# chmod 755 admin.bash
[root@ip-172-31-33-47 scripts]# cat admin.bash
#!/bin/bash

top | head -10
df -h
du -k
hostname
iostat

[root@ip-172-31-33-47 scripts]# ./admin.bash
top - 10:27:16 up 35 min, 2 users, load average: 0.00, 0.00, 0.00
Tasks: 109 total, 1 running, 108 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 3.1 sy, 0.0 ni, 96.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 912.9 total, 510.3 free, 166.5 used, 236.1 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 614.7 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0 173220 17536 10900  S   0.0   1.9   0:01.03 systemd
    2 root        20   0        0        0        0  S   0.0   0.0   0:00.00 kthreadd
    3 root        20   0        0        0        0  S   0.0   0.0   0:00.00 pool_workqueue_release

Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        420M   0  420M   0% /dev
tmpfs           457M   0  457M   0% /dev/shm
tmpfs           183M 440K  183M   1% /run
efivarfs        128K 2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1  8.0G 1.6G  6.4G  21% /
tmpfs           457M   0  457M   0% /tmp
/dev/nvme0n1p128 10M 1.3M  8.7M  13% /boot/efi
tmpfs           92M   0   92M   0% /run/user/1000

ip-172-31-33-47.ec2.internal
Linux 6.18.35-68.129.amzn2023.x86_64 (ip-172-31-33-47.ec2.internal)    07/03/26    _x86_64_    (2 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.37    0.00    0.15    0.05    0.04   99.39

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
nvme0n1             3.21         73.13         81.24          0.00     154607     171768         0

```

5) Created and ran a bash script to change the file name using input from the user.

```

root@ip-172-31-33-47:/salman/scripts
[root@ip-172-31-33-47 scripts]# cat task1.bash
#!/bin/bash
#This script will rename a file

echo Enter the file to be renamed
read oldfilename

echo Enter the new file name
read newfilename

mv $oldfilename $newfilename

echo The file has been renamed as $newfilename

[root@ip-172-31-33-47 scripts]# ./task1.bash
Enter the file to be renamed
task1
Enter the new file name
task2
mv: cannot stat 'task1': No such file or directory
The file has been renamed as task2
[root@ip-172-31-33-47 scripts]#

```

6) Created and ran a script to show the current date, time and user data.

```
root@ip-172-31-33-47:/salman/scripts
[root@ip-172-31-33-47 scripts]# vi case.bash
[root@ip-172-31-33-47 scripts]# chmod 755 case.bash
[root@ip-172-31-33-47 scripts]# cat case.bash
#!/bin/bash
echo
echo Please choose one of the options below
echo
echo 'a = Display date and time'
echo 'b = List file and directories'
echo 'c = List users logged in'
echo 'd = Check system uptime'
echo
read choices
case $choices in
a) date;;
b) ls;;
c) who;;
d) uptime;;
*) echo Invalid choice - Bye.
esac
[root@ip-172-31-33-47 scripts]# ./case.bash
Please choose one of the options below
a = Display date and time
b = List file and directories
c = List users logged in
d = Check system uptime
a
Sat Jul 4 10:31:58 UTC 2026
[root@ip-172-31-33-47 scripts]# ./case.bash
Please choose one of the options below
a = Display date and time
b = List file and directories
c = List users logged in
d = Check system uptime
c
ec2-user pts/0          2026-07-04 10:27 (106.192.21.3)
ec2-user pts/1          2026-07-04 10:27 (106.192.21.3)
[root@ip-172-31-33-47 scripts]# |
```

7) Created and ran a bash script to backup files depending on the days of the week.

```
root@ip-172-31-33-47:/salman/scripts
[root@ip-172-31-33-47 scripts]# vi backup.bash
[root@ip-172-31-33-47 scripts]# chmod 755 backup.bash
[root@ip-172-31-33-47 scripts]# cat backup.bash
#!/bin/bash
NOW=$(date +"%a")
case $NOW in
Mon)
echo "Full backup";;
Tue|Wed|Thu|Fri)
echo "Partial backup";;
Sat|Sun)
echo "No backup";;
*) ;;
esac
[root@ip-172-31-33-47 scripts]# ./backup.bash
No backup
[root@ip-172-31-33-47 scripts]# |
```